

# KAJAL KUMARI

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## Education

### Ramgarh Engineering College

B.Tech in Computer Science Engineering; CGPA: 7.14/10

Ramgarh, India

May 2027

### Dhanbad Public School

12th Standard; 80.6%

Dhanbad, Jharkhand

May 2023

### Dhanbad Public School

10th Standard; 93.2%

Dhanbad, Jharkhand

August 2021

## Technical Skills

**Core:** Machine Learning, Object-Oriented Programming (OOP), Data Structures & Algorithms, Computer Networks, DBMS, Operating Systems

**Languages:** C++, Java, Python, SQL

**Web:** HTML, CSS

**Certificates:** : JAVA DEVELOPMENT (InternPe)

## Work Experience

### Indian Institute of Technology (ISM) Dhanbad

Summer Research Intern

Onsite

May 2025-July 2025

- Built a **Python-based Federated Learning model** for a Diabetes Detection System using a neural network architecture that achieved **92% accuracy** in predicting patient diabetes status while ensuring full data privacy.
- Implemented data preprocessing for critical medical attributes (glucose, BMI, blood pressure, age) and the FedAvg algorithm, improving data privacy by **100%** and model integrity by **100%** over centralized baseline models.
- Trained on distributed medical data across multiple clients and achieved consistent prediction reliability across heterogeneous datasets.
- Integrated a Blockchain-based logging system for secure model updates with transaction finality of **5 seconds**, ensuring transparency and trust in healthcare AI systems.

## Projects

### Phishing Websites Detection (Machine Learning & Web development )

May 2025

- Developed a website powered by machine learning to enhance security by 70% and reduce phishing attacks 50%.
- Implemented machine learning classification models with a 97.4% accuracy rate in identifying phishing websites .
- Performed comprehensive feature extraction from website content and metadata utilizing advanced data processing techniques to enhance feature selection resulting in a 25% improvement in model accuracy.
- Project repository: [github.com/Phishing-Detection](https://github.com/KajalKumari985/Phishing-Detection).

### Fetal Health Prediction system (Machine Learning & Web development )

November 2024

- Developed an advanced Machine Learning model combined with a website in Python to predict fetal health, transformative prenatal care model increased early diagnosis accuracy by 88% and reduced false negatives by 45%.
- Contributed to frontend and backend development, enhancing project contributions by 50% through comprehensive involvement.
- Project repository: [github.com/Fetal-Health-Detection](https://github.com/KajalKumari985/Fetal-Health-Detection).

## HONORS and AWARDS

Participated in Web Conference of Digital Start-up Hack A Fest - REC:

December 2024

Qualified for Smart India Hackathon (SIH) 2024

September 2024

## Leadership and Volunteering

### Robotics Club: Technical Team - Ramgarh Engineering College

October 2023 - Present

- Organized 3 workshops and 2 technical sessions with 150+ participants, developing practical robotics projects to enhance hands-on technical skills.
- Facilitated weekly club meetings and initiatives with 40+ members to coordinate project development and knowledge sharing in robotics.

### National Service Scheme (NSS) - REC

January 2024 - Present

- Organized field survey and quantitative analysis to understand villager's issues and conducted 3 camps with 200+ students and 3 village participation, improving community engagement and leading to increased awareness.
- Organized and led 10 educational programs for underprivileged children, enhancing literacy and basic education for over 150 children in rural areas, resulting in a 40% increase in reading proficiency.